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Heterogeneous Multi-Layer Constellation for Future Satellite Internet: Framework, Optimization, and Evolution Vision

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Abstract

The alternative strategy for future space-based satellite (SAT) internet development involves a potential transition from single-layer low Earth orbit (LEO) system to sophisticated heterogeneous multi-layer constellations (HMLCs). HMLC integrates SATs from different orbital layers, including very low Earth orbit, LEO, medium Earth orbit, and geostationary orbit, potentially incorporating communication, navigation, and remote sensing (CNR) capabilities, with the ability to enhance coverage, communication capacity, geometric dilution of precision, resilience, resolution, *etc.*, then offering the diversification of service functions. Realizing this potential hinges on multi-SAT collaboration driven by key technologies including inter-satellite link, advanced antenna technology, on-board processing, topology-aware routing, and artificial intelligence. Designing these sophisticated systems inevitably involves multi-objective optimization problem (MOP) to trade-off conflicting performance metrics, especially in optimizing integrated CNR services. In addition, HMLC's structural flexibility and integration capabilities also provide an important foundation for future mission configurations extending into deep space. This article explores the HMLC framework, enabling technologies, and the MOP paradigm. Furthermore, it outlines a vision towards intelligent and agile HMLCs, emphasizing future directions in rapid constellation assessment using surrogate models and dynamic in-orbit adaptation to meet evolving demands, paving the way for building adaptive and efficient intelligent space-based networks to support future deep space missions such as Earth-Moon collaboration.

I. Introduction

The skies above are becoming increasingly populated, not just with stars, but with constellations (CONs) of human ingenuity. The palpable excitement surrounding mega-CONs, mainly in low Earth orbit (LEO) satellite (SAT) space, spearheaded by ambitious projects like *GuoWang* (GW), G60, Starlink, Kuiper, OneWeb, Iridium Next, *etc.*, signals a transformative era for global connectivity [1]. These orbiting networks promise to bridge the digital divide, delivering broadband internet to the most remote corners of the globe. This surge in SAT deployment aligns perfectly with the grand vision of future sixth generation (6G) networks [2], often conceptualized as space-air-ground-sea integrated network (SAGSIN) [3]. Within SAGSIN, SATs transcend their traditional role as mere relays, evolving into integral, intelligent nodes within a unified communication fabric, fundamentally reshaping how we connect and share information globally.

However, accompanying the focus on research into fifth generation (5G) service application architectures with SAT access support [4], the limitations of relying solely on homogeneous, single-layer LEO CONs, or on large-scale mega-CON strategies that emphasize SAT quantity rather than architectural diversity and intelligent optimization, are becoming increasingly apparent through our ongoing exploration of space-based SAT internet. The latter remains a distant prospect for most institutions such as those in developing countries [5], where the development of CON deployment is still in its initial stages, while single-layer architecture with inter-SAT link (ISL) is also unable to effectively meet the diverse quality of service (QoS) requirements of modern applications [6], [7]. High-throughput data transmission, ultra-reliable low-latency communications, precise navigation and timing, and high-resolution Earth observation all present distinct requirements for future integrated CON network systems, challenging the ability for simultaneous optimization within single orbital regime [8], [9]. For instance, a monolithic LEO network often struggles to efficiently serve a diverse range of users, from densely populated urban centers and underserved rural communities to aircraft traversing oceans and ships navigating remote waters [5]. In addition, the commonly utilized LEO space is nearly saturated, and as shown in Fig. 1, currently it contains a large amount of space junk with collision risk.

To overcome these challenges and develop space-based infrastructure with the potential for

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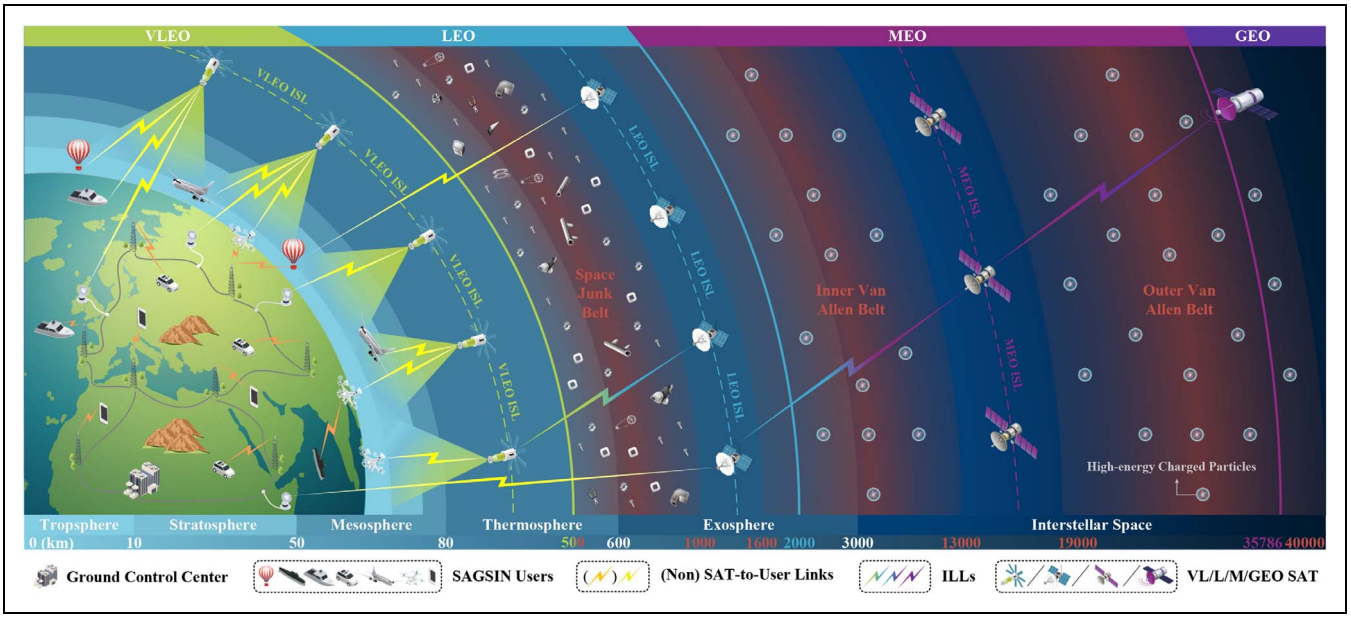


FIG. 1. ISL/ILL-enabled HMLC-centric framework of SAGSIN and an illustration of the coordinated services for space, air, ground, and sea user segments.

integrated *Earth-near Earth-deep space* services, the architectural evolution direction is advocated toward heterogeneous multi-layer CON (HMLC) [7], which is also expected to serve as the foundational architecture for achieving a more integrated space internet and provide structural possibilities for future CON systems to support services such as lunar orbit communications and deep space exploration missions. HMLC represents a necessary evolution that constructs a space-based network by integrating SATs across multiple orbital layers, potentially spanning very LEO (VLEO) [5], LEO, medium Earth orbit (MEO), geostationary Earth orbit (GEO), and future extended deep space relay layers. Crucially, this heterogeneity exists beyond orbital altitude heterogeneity and also encompasses functional heterogeneity. HMLC can accommodate SATs dedicated to specific mission functions [6] such as communication, navigation, and remote sensing (CNR), or increasingly, integrated platforms capable of performing multiple functions concurrently [10]. HMLCs constitute the spatial component of SAGSIN, characterized by SAT layers interconnected via inter-layer links (ILLs) as illustrated in Fig. 1, serving a multitude of users across terrestrial, aerial, and maritime domains [11]. Its core advantage lies in the potential for intertwined collaboration between layers, resulting in a highly resilient, adaptable, and highly capable spatial infrastructure far surpassing the individual sum of its parts. This collaborative framework promises unprecedented levels of performance, resilience, and adaptability.

Moreover, with the continuous advancement of deep space exploration missions such as lunar and Mars exploration, the construction of CON networks with Earth-Moon integrated and Earth-Mars integrated capabilities has gradually become an important direction for the evolution of future space infrastructure. Fig. 2 shows an illustrative example of the vision for extending CON networks to the Moon. Under this trend, HMLC can not only support service integration in near-Earth region, but is also expected to have a natural ability to evolve toward deep space. For instance, by expanding the deployment of a deep space relay layer on top of the GEO base layer, MEO

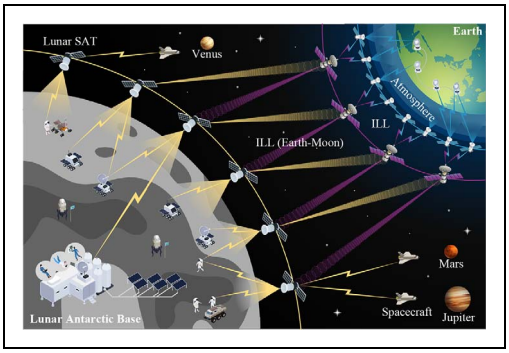


FIG. 2. Earth-Moon integrated HMLC for deep space exploration extension.

support layer, and LEO high-throughput layer architecture, it is achievable to support the continuity of the Earth-Moon-Mars communication link, catering to data relay across billions of kilometers and cross-domain resource coordination.

This article navigates the insightful landscape of HMLCs, focusing on the critical interplay between architectural frameworks, enabling technologies, and advanced design methodologies. We delve into the structure of HMLC, emphasizing the synergies between layers and the growing importance of integrated CNR (ICNR) services [10]. We consider dissecting the key technologies that promote multi-SAT collaboration, enable sophisticated optimization, and naturally evolve toward future deep space scenarios. A pivotal theme is the involvement of multi-objective optimization problem (MOP) and its design [12], which are essential for balancing the inherently conflicting demands embedded in HMLC framework, especially in integrated systems [6], [10]. In addition, it is expected to explore future artificial intelligence (AI)-enabled agile HMLC vision, such as rapid assessment capabilities and dynamic adaptation strategies, which also supports the long-term evolution of future deep space missions like Earth-Moon coordination. The discussion will draw on recent advances [1], [5], [6], [7], [8], [9], [12], [13], [14], [15] in CON design,

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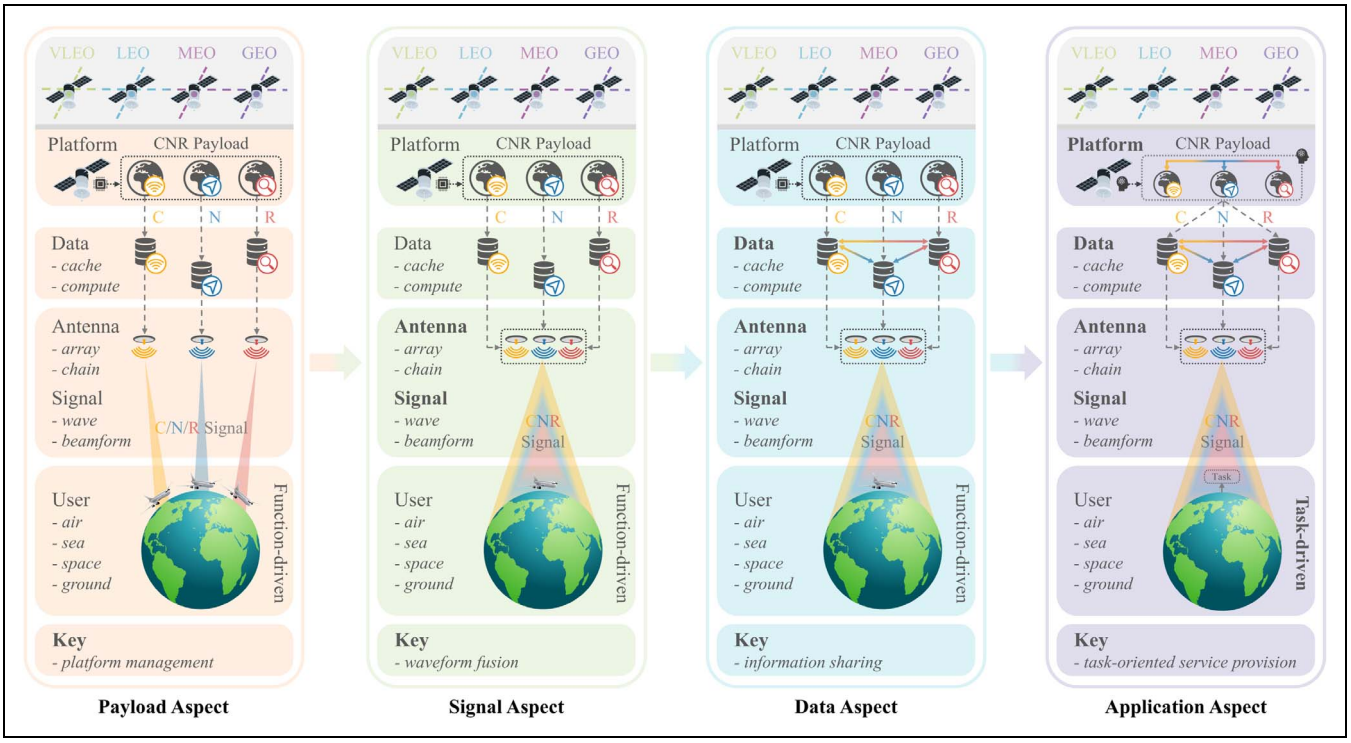


FIG. 3. Functional capability heterogeneity in the HMLC framework and the ICNR integration mechanism under different intensity stages.

optimization, routing, service integration, *etc.*, aiming to provide insights for researchers and engineers shaping the future space-based SAT internet.

II. HMLC Framework and the Heterogeneity

As mentioned above, the HMLC framework incorporates heterogeneity in terms of CON's physical architecture deployment (Fig. 1) and platform integration capabilities (Fig. 3), supporting effective synergies and performance enhancements.

A. Architecture Aspect: Strengths and Synergies

Compared with single-layer CONs, HMLC's motivation stems directly from the diversity of service requirements and the necessity of releasing synergistic advantages. By strategically distributing SAT assets across different orbital altitudes and equipping them with diversified and integrated functions, HMLCs offer a more robust, efficient, and adaptable solution. Combining the inherent low latency of LEO and even VLEO SATs with the expansive coverage footprints of MEO and GEO SATs, optimized services can be tailored to specific demands. In addition, as mentioned in Fig. 1, the escalating issue of orbital debris and saturation of prime LEO positions make it advisable to allocate assets to less congested altitudes for enhanced long-term sustainability and resilience. Different layers naturally lend themselves to different primary services: *i)* despite challenges like atmospheric drag, VLEO offers potential for exceptionally high-resolution remote sensing or ultra-low-latency regional communications [5]; *ii)* LEO remains the mainstay for broadband internet access [2]; *iii)* MEO provides a stable platform favored for navigation owing to its long visibility, stable geometry, and legacy, although LEO-based navigation augmentations are increasingly proposed; *iv)* GEO excels at wide-area broadcast and data relay.

The HMLC architecture avoids being constrained by the limitations of any single layer, instead leveraging their complementary strengths. For instance, latency-sensitive interactive data can be routed through LEO/VLEO layers [13], while delay-tolerant broadcast traffic can utilize GEO, and navigation services primarily rely on MEO. While each orbital layer exhibits distinct latency-coverage trade-offs, cross-layer coordination enables partial functional substitution, enhancing resilience and service continuity. More significantly, the diversity of this cross-orbit architecture provides natural service extension capabilities for expansion into deep space scenarios such as the Earth-Moon and Earth-Mars orbits. Specifically, high-orbit assets like MEO/GEO layer can serve as the first-hop access layer for future deep space communication links.

B. Service Aspect: CNR Integration

Beyond the physical distribution across layers, the functional dimension of HMLCs is equally critical. Current system design is shifting away from single-service SAT platforms toward a multifunctional, integrated system paradigm, which incorporates CNR capabilities into a single CON and provides unified support for ground and space missions [10].

The integration of CNR functionalities within an HMLC framework offers compelling advantages over deploying separate, stove-piped systems. On the one hand, reuse can be achieved at multiple levels, including spectrum and payload, greatly improving the overall utilization efficiency. On the other hand, performance can be enhanced through the cross-integration of data between functions[1]. Of course, this tightly coupled integration also raises challenges such as increased coordination complexity, greater on-board computing and power demands, and more strict scheduling optimization. As shown in Fig. 3, integration intensity can be divided into four stages:

[1] Navigation signals assist communication systems with beam steering and interference mitigation; communication links can propagate real-time remote sensing data or differential corrections to improve navigation accuracy; remote sensing data can provide information for communication routing decisions or valuable background information for navigation applications.

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i) *Payload aspect*. The weakest integration, where a single SAT can provide CNR functionality but operates almost independently. Except for basic SAT spatial information including six elements, direction, and attitude, which are shared, each function utilizes its own dedicated hardware to execute its mission. Key to payload aspect is platform management;

ii) *Signal aspect*. CNR signals are jointly designed to simultaneously carry communication, navigation, and remote sensing capabilities, enabling more efficient utilization of limited spectrum resources. Key to signal aspect is waveform fusion, with a simplified option being trade-offs in time, frequency, or spatial dimensions, *i.e.* orthogonal allocation of wireless resources to support different functions;

iii) *Data aspect*. It supports information sharing between different functions, enabling cross-functional data coordination and network optimization. Combining footnote[1] and taking autonomous driving and as an example (other integrated ground observation, communication, and navigation missions have similars), remote sensing results can reveal spatial distribution, providing support for traffic-aware communication and navigation resource scheduling; navigation information such as location and speed can assist in predicting communication channels, enhancing the adaptability of transceivers; and by analyzing large-scale channel state information (CSI) data, the regional channel environment can be inferred to optimize remote sensing beamforming and enable location-based CSI fingerprinting. Key to the data aspect is information sharing;

iv) *Application aspect*. Centering on task objectives, regard communication, navigation, and remote sensing functions as interchangeable and collaborative service resources, and achieve the optimal combination under dynamic system conditions. Distinct from the first three stages of integration, which satisfy predefined functional performance requirements, application integration focuses on how to accomplish specific tasks. Key to application aspect is task-oriented service provision, which relies on service demand domain modeling, network status evaluation, and intelligent global optimization, reflecting a paradigm shift from *function-driven* to *task-driven*.

In comparison with conventional dual-function integration schemes typically limited to dual-service coupling within single-orbit architectures, the proposed ICNR mechanism within the HMLC framework achieves cross-layer, cross-service, and cross-domain collaboration. This multi-stage integration not only enhances spectrum and payload reuse but also enables dynamic service interchangeability and autonomous reconfiguration, offering significantly higher adaptability for multi-mission operations and future deep space extension.

III. Key Technologies for Multi-SAT Collaboration in HMLC

The successful operation and optimization of HMLC relies on synergistic integration of advanced technologies. As shown in Fig. 4, these driving factors not only promote collaboration among multi-layer SAT assets, but also lay the foundation for future expansion into challenging scenarios such as disaster response or deep space services. Expectedly, they allow HMLC to gradually evolve into a comprehensive system covering a continuous domain from near-Earth to deep space, where each technology must be refined to sustain operations under long-delay, high-radiation, and limited-support conditions.

A. ISL and ILL: Weaving SAT Collaboration Mesh

The cornerstone of HMLC collaboration is ISL and ILL, which serve as data highway in space across all SATs. In particular, as shown in Fig. 2, ILL is one of the key technologies for the natural evolution of future deep space scenarios.

By establishing direct communication links between SATs, the reliance on ground station relays is significantly reduced, and end-to-end communication latency is greatly lowered, making it particularly suitable for long-distance links that require multiple SAT hops. This feature is crucial not only for marine and polar regions but also for deep space relays of Earth-Moon and Earth-Mars, where ultra-long optical paths and severe link fading require advanced acquisition, pointing, and tracking mechanisms together with adaptive coding schemes to guarantee reliability.

In terms of category, while traditional microwave ISL has been in use, its development trend is rapidly shifting toward optical ISL (OISL), which imposes lower requirements on terminal size, weight, and power consumption, while significantly enhancing bandwidth potential, interference resistance, and security [11]. For deep space extension, hybrid optical ILL architectures can provide redundancy and enable sustained inter-planetary connectivity despite long propagation delays. Certainly, OISLs pose considerable technical challenges, primarily in terms of the precise acquisition, pointing, and tracking required to maintain links between rapidly moving and distant SATs. However, it is promising that robust ISL/ILLs topology designs can ensure connectivity even in the face of dynamic SAT movements and potential node failures, making them a key component of resilient HMLC designs.

B. Advanced Antenna Technologies: Enhancing Beam Agility

Effective communication between SATs and the ground, as well as between SATs themselves, relies heavily on advanced antenna technologies. Phased array antennas (PAAs), *i.e.* electronically steerable antennas (ESAs), are becoming indispensable for both SAT payloads and ground terminals. Unlike traditional mechanically steered dishes, PAA/ESA can electronically steer the beam almost instantaneously. This enables multi-beam operation, whereby a single SAT can provide dedicated beam services to multiple users or regions simultaneously, greatly increasing system capacity. Rapid beam handover is critical for maintaining links with high-speed LEO SATs and ensuring seamless handovers between them [4].

For deep space scenarios, PAAs must maintain high-gain narrow beams with extremely precise pointing control and thermal stability to withstand intense temperature variation and radiation exposure. The adoption of adaptive beam-forming with predictive attitude compensation further ensures long-range link

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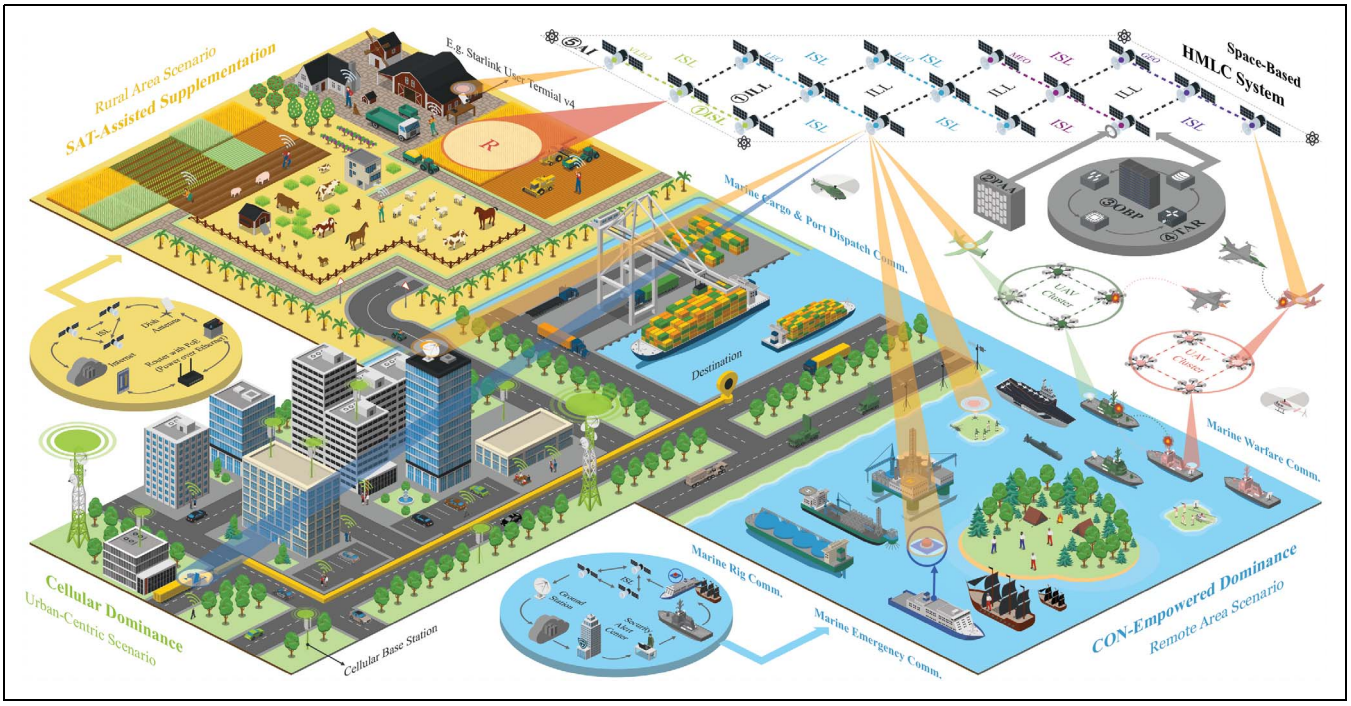


FIG. 4. Various typical application scenarios involving CNR services in intelligent space-based HMLC system and its key supporting technologies (① ∼ ⑤).

continuity between planetary nodes. In addition, precise beam forming and effective power concentration can help alleviate interference, which is critical in the dense signal environment of HMLCs and for enabling spectrum sharing between SAT and terrestrial systems.

C. Regenerative SAT and OBP: Enabling Autonomy

Complementing the connectivity provided by ISL is the intelligence enabled by on-board processing (OBP). Traditional bent-pipe SATs function solely as a *mirror*, amplifying radio signals and relaying them between ground stations without modifications. While regenerative SAT equipped with OBP can directly perform multiple processing tasks such as modulation, demodulation, encoding, decoding, *etc.* on the board.

OBP-enabled routing capabilities enable SAT packets to traverse ISL networks intelligently and efficiently, minimizing command intervention from ground at each hop. It facilitates intelligent resource allocation, enabling SAT to dynamically manage bandwidth and power based on real-time traffic demands and link conditions. Security can also be enhanced through onboard encryption and decryption. Crucially, OBP enables SAT to collaborate directly and process data from multiple sources, laying the foundation for a genuine intelligent space network. Under deep space conditions, OBP must incorporate radiation-hardened processors, autonomous fault-recovery logic, and delay-tolerant control mechanisms to operate with minimal Earth supervision, effectively transforming each SAT into an independent decision-making node.

D. Topology-Aware Routing: Ensuring Efficient Data Flow

In the temporal and dynamic environment of HMLCs, topology-aware routing (TAR) plays a vital role in enabling efficient data delivery by leveraging the predictable yet constantly evolving nature of SAT orbits. Unlike terrestrial networks with irregular topological changes, SAT movements follow deterministic trajectories. TAR systems leverage this to predict upcoming changes in link availability and proactively adjust routing paths to minimize latency and packet loss. For deep space extensions where link latency can span several minutes, TAR mechanisms must incorporate predictive scheduling and delay-tolerant routing protocols to ensure data integrity despite intermittent connectivity. These systems preserve physical-logical topology alignment, enabling real-time routing updates from orbital dynamics.

In multi-layer CON network systems, TAR needs to balance route selection between higher and lower layers, dynamically optimizing data flow through indicators such as link quality, congestion, and energy availability. Advanced designs treat each layer as a routing domain, using hierarchical protocols and inter-domain coordination to handle cross-layer traffic.

E. AI Integration: Empowering Intelligence and Adaptation

Considering the future scale and functional enhancements of HMLC, AI technology is the foundation of its autonomous management. At the on-board level, lightweight inference models leverage OBP resources for real-time decision-making; at the network level, cooperative AI agents utilize ISL-enabled information sharing to achieve coordinated optimization among SATs. It can efficiently process massive amounts of data for fault prediction, route optimization, resource scheduling, and dynamic response to mission and traffic variations. For example, research has shown that deep reinforcement learning excels in optimizing long-term benefits across multiple objectives such as beam control, spectrum allocation, and traffic routing; federated learning enables SATs to train models locally and exchange parameter

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updates, reducing communication overhead while enhancing the system's overall intelligence, particularly suited for anomaly detection and predictive maintenance tasks.

In the context of deep space expansion, AI plays a pivotal role in enabling in-situ decision-making, anomaly correction, and autonomous task reconfiguration without real-time ground control. Distributed learning and on-orbit inference further allow HMLCs to sustain performance under high latency, communication disruptions, and unpredictable environments. Further integration of OBP and ISL enables CON-level on-orbit inference and collaboration, granting the system self-awareness and self-optimization capabilities, and will be vital elements supporting flexible and resilient HMLC framework.

IV. MOP CON Design and Optimization Paradigm

To fully realize the potential of space-based HMLC systems, design considerations need to go beyond single metrics and embrace a comprehensive and superior Pareto MOP optimization paradigm that balances performance, cost, and resilience across a variety of mission scenarios. In particular, for the future integrated *Earth-near Earth-deep space* service system, optimization objectives need to take into account traditional communication missions while also meeting multiple requirements such as deep space link extension, functional redundancy, and intelligent scheduling. Meanwhile, for the implicit specificity of CON design objectives, we suggest compensating with the high-fidelity CON system establishment.

A. The Complexity of HMLC Design Objectives

The objectives driving HMLC design are diverse and sophisticated, which can also be illustrated by the multiple services involved in Fig. 4. The key performance metrics of traditional communications CONs could include global or regional coverage and throughput (customized based on real traffic analysis [5]), while ensuring end-to-end latency and network resilience. The system's ability in recovering from various disturbances, *e.g.* multi-SAT signals and SAT failures, is also critical for certain missions. While for the integrated CNR CON design supporting the coordination of ground, airspace, and deep space missions, optimizing navigation accuracy, *e.g.* minimizing geometric dilution of precision (GDOP) [12], and remote sensing performance, *e.g.* maximizing spatial resolution or revisit frequency [10], become equally important objectives. All performance objectives shall be achieved under strict constraints, primarily including SAT deployment costs and on-orbit power consumption.

Typically, considering multiple objectives means trade-offs. For illustration, enhancing navigation accuracy may necessitate specific orbital configurations or additional SATs, potentially increasing costs and latency, or slightly reducing communication coverage in certain regions; achieving permanent ISL connectivity to enhance resilience may require denser CONs or specific phase divisions, thereby impacting costs and complexity. Understanding and navigating these inherent trade-offs is central to the effective design of HMLC, and we advocate the Pareto MOP scheme, which offers a more comprehensive and adaptable solution analysis compared to weighted model and ϵ-constrained model.

B. Preferable Pareto MOP Scheme and Novel Configurations

Following the previous section, an advanced Pareto framework is advocated in the context of multi-objective design and optimization for HMLC, in order to avoid the limited insights of single-objective or simple models, explore high-dimensional design spaces, and identify a series of optimal trade-off solutions, *i.e.* the Pareto front. Heuristic algorithms such as particle swarm optimization (PSO) and genetic algorithm (GA), including non-dominated sorting genetic algorithm-II (NSGA-II) and its variants such as NSGA-III have been highly validated in this field for their effectiveness and convenience [5], [7], [12]. As shown in Fig. 5, these algorithms operate on a population of candidate CONs and iteratively optimize them through simulated evolution to converge toward the Pareto front, where the high-fidelity CON evaluation system is necessary with respect to the implicit specificity of the fitness involved in this field (refer to Section IV-C).

Considering that future space services will extend from the ground to the moon and beyond, structural optimization needs to support multi-stage, scalable, and evolutionary configurations. For example, LEO-MEO CONs can be deployed initially to cover near-Earth missions, and then evolve to integrate GEO/MEO gateways or space relay CONs to support closed-loop communications in deep space. In addition, the pursuit of optimal HMLC promotes the exploration of novel configurations beyond tradition such as Walker or Flower. For instance, the hybrid inclination (HyInc) Walker we proposed in [12], heuristically relaxes the uniform inclination constraint across all orbital planes based on the lunar integrated communication and navigation CON demands and combined with Earth-Moon differences, providing greater mission adaptability and MOP trade-off space. Notably, configuration exploration combined with specific mission-oriented optimization can leverage significant improvement in MOP scenarios, particularly for designs prioritizing overall integration or balancing challenging targets such as resilience and deep space extension.

C. Implicit Fitness and High-Fidelity Integration Architecture

One of the core challenges in HMLC design lies in the implicit fitness represented by non-analytic expressions. This characteristic imposes higher demands on computational resources, and although it results in higher operational complexity, CON evaluation still relies on an integrated solution architecture that combines high-fidelity CON evaluation systems with external heuristic optimization engines to achieve closed-loop feedback from design variables to performance metrics. Specifically, the high-fidelity evaluation system provides multi-dimensional performance feedback (*e.g.*, coverage, GDOP, and resolution), while heuristic algorithms such as PSO and GA update the design variables accordingly, forming an iterative feedback loop rather than two independent processes. The architecture highlights partial decoupling and deep collaboration between internal and external modules, guiding heuristic algorithms such as PSO and EA to intelligently search in CON diversity space through continuous feedback. Compared to pure mathematical analytical solutions with lower complexity, the integrated architecture

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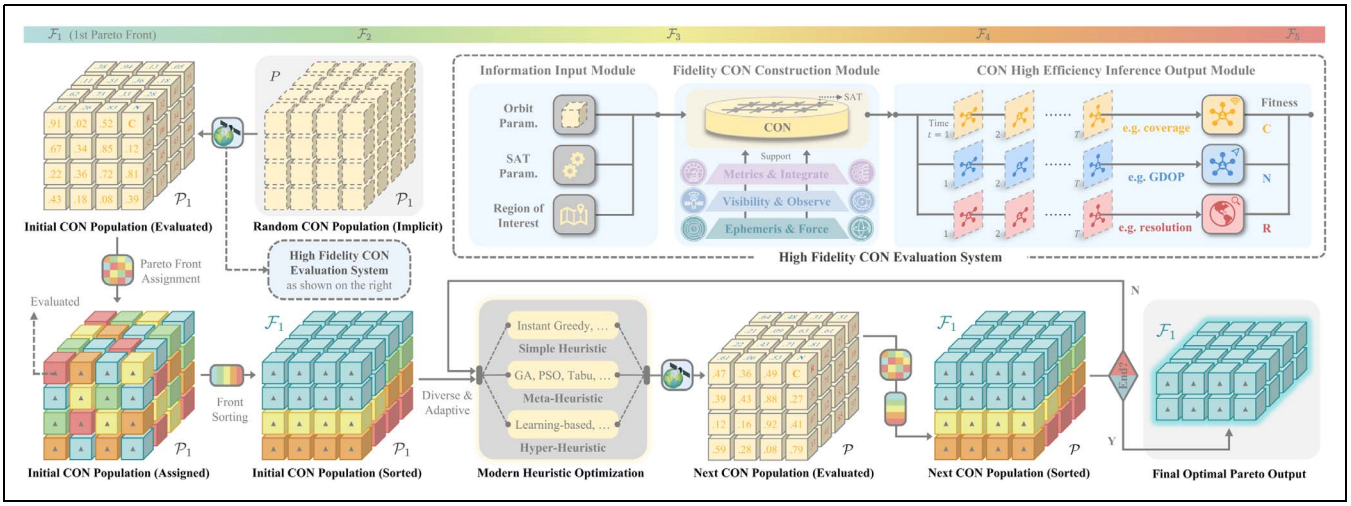


FIG. 5. Integration of population-based heuristic design in HMLC and the CNR optimization paradigm (normalized fitness and P denotes population size).

provides stronger practical guidance for CON deployment. The integrated architecture has strong scalability and universal adaptability, and is suitable for structural migration in future cross-domain heterogeneous mission scenarios such as air, space, and the Moon.

As shown in Fig. 5, the high-fidelity CON evaluation system serves as the internal fitness engine of the entire architecture, supporting modular evaluation of metrics such as coverage, GDOP, and resolution. Each module can be loosely coupled, while retaining critical dependencies, to facilitate task adaptation and structural expansion through component-level modifications without restructuring the entire system. By leveraging the synergy between high-fidelity evaluation and MOP algorithms, the architecture can systematically explore the CON Pareto front. Under the premise that the multi-objective trade-off relationship is not explicitly solvable, it can effectively identify the optimal Pareto solutions around inflection points among structural performance by combining various visualization approaches. Besides, considering the modular decoupling design, other advanced search algorithms or neural network surrogate models can be seamlessly integrated for rapid adaptation to other scenarios [15]. This architecture serves as a scalable paradigm with high practical value for future multi-task and multi-strategy HMLC design.

V. Vision and Future Research Directions

While MOP paradigm effectively enables integrated optimization of HMLC design, operational realities are far more dynamic. Traffic patterns fluctuate, mission priorities evolve, and SATs may degrade or fail. To maintain optimal performance and resilience throughout the lifespan, future HMLCs need to go beyond static design and embrace agility, driven by rapid performance evaluation and intelligent adaptability. Especially with the emergence of missions to the Moon and farther into deep space, CONs need to have the potential for dynamic expansion and flexible evolution.

A. Rapid CON Evaluation and Optimization Design

One of the main bottlenecks in the initial CON design and its continuous operation management is the computational cost of high-fidelity simulation evaluation. Running complex simulations in numerous scenarios to analyze coverage, latency, connectivity, or interference can take hours or days, hindering rapid design iterations. A critical future direction, consistent with ongoing research efforts, includes the development of rapid CON evaluation models. This involves machine learning-enabled surrogate models, essentially approximations of the efficiency of complex evaluation systems. Techniques such as ensemble learning and neural networks can be utilized on large datasets from high-fidelity simulation systems. Once trained and validated, surrogate models can provide faster predictions based on CON parameters or environmental conditions, thereby supporting rapid analysis and structural design for complex objectives such as deep space link modeling and dynamic relay node layout, greatly reducing CON simulation time from days to hours basically and accelerating the iterative processes of optimization paradigm proposed in Section IV. The relevant schematic diagram is shown in Fig. 6(a).

B. Dynamic CON Adaptation and Reconfiguration

The ability of rapid performance evaluation unlocks the potential for dynamic CON adaptation. If the network can quickly learn how its performance is affected by changeable conditions, or how it will be affected by potential changes, then implementing related adjustments becomes a viable strategy. This vision requires SATs with enhanced maneuverability, possibly relying on advanced, highly efficient electric propulsion systems to perform more frequent and vital orbital modifications than currently typical situations (*e.g.* phase changes, altitude corrections, and inclination shifts). Combining rapid evaluation models with enhanced maneuverability enables agile on-orbit adjustment. Examples of these possible scenarios: *i)* HMLC automatically detects coverage shortfalls due to unexpected demands; *ii)* rapid evaluation models confirm the issue and assess potential adjustments; *iii)* AI-controlled agent commands specific SATs to adjust altitude or inclination to resolve the shortfall optimally, all within operational timeframe. The relevant schematic diagram is shown in Fig. 6(b). Additionally, reconfiguration cadence is related to propulsion capabilities and mission scheduling cycles, where beam or routing reassignments can be accomplished in seconds to minutes, while orbital element

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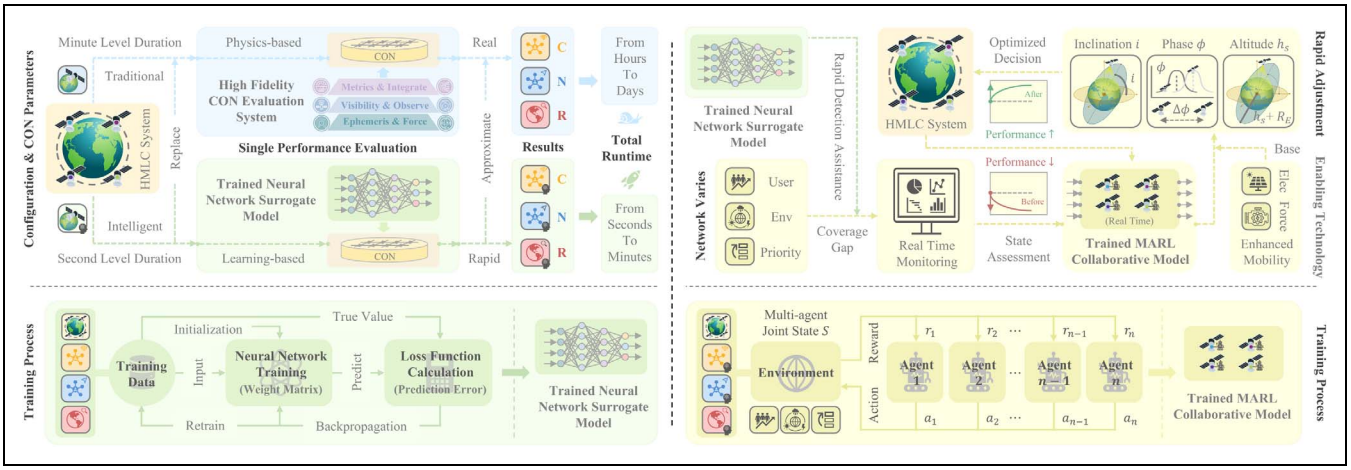


FIG. 6. Diagram of vision and future directions. (a) Left: rapid CON evaluation and optimization design; (b) Right: dynamic CON adaptation and reconfiguration.

updates can take hours to days. And predictive AI planning enables pre-scheduled redundancy during scheduled downtimes.

This is expected to be integrated into multi-agent reinforcement learning (MARL), where the model is trained to adjust CON based on real-time feedback from fast evaluation agents and network sensors, enabling CON to autonomously adapt to dynamic traffic demands, fluctuating environment, or changing task priority (e.g. shifting CNR resources to remote areas). When working on missions like deep space node relay scheduling and Earth-Moon link maintenance, this dynamic self-evolution capability will be one of the essential support capabilities for cross-domain ICNR platforms. In practice, this agility allows HMLCs to maintain service continuity under varying conditions and mission loads, ensuring that future space networks remain responsive and resilient even across deep space operations.

C. The Future: AI-Native and Self-Optimizing HMLCs

Extrapolating these trends leads to the ultimate vision of HMLCs as fully integrated, AI-native systems. In this future paradigm, artificial intelligence is not merely tools applied to specific tasks but form the core intelligence fabric orchestrating the entire network. That is, it will consider global agent and comprehensively manage multi-layer and cross-functional resource allocation. And it will enable deep collaboration between CNR functions on multi-layer SATs, seamlessly integrate data, and provide integrated services. And incorporating predictive intelligence, where AI agents anticipate future CON states and adjust optimization objectives accordingly, will be an essential evolution of the proposed AI-native paradigm.

In addition, AI-driven active fault detection, prediction, and autonomous recovery mechanisms will ensure unprecedented system resilience. Autonomous navigation and maneuvering will become standard practice, continuously optimizing orbital performance, lifespan, and collision avoidance. Especially as CONs expand further into deep space, such as the Moon and the Mars, AI's native capabilities will support autonomous decision-making and closed-loop control in conditions where ground control is lacking and latency is high, truly realizing an intelligent space network. Furthermore, the network itself will possess the ability to learn and evolve throughout its operational lifespan, consistently adjusting its configuration and strategies in response to evolving objectives and dynamic environment. Ultimately, AI-native HMLC will seamlessly integrate with future ground-based 6G networks, forming a truly unified global CNR infrastructure. This AI-native autonomy represents the cornerstone of next-generation space systems, transforming SAT networks from conventionally pre-scripted infrastructures into self-evolving intelligent ecosystems capable of supporting interplanetary-scale missions.

VI. Conclusion

In conclusion, HMLCs represent a paradigm shift beyond homogeneous single-layer systems, integrating diverse orbital assets and CNR functionalities to address increasingly complex space-based service demands and construct network architectures with long-term service capabilities and evolutionary flexibility. This architectural evolution, enabled by ISL, advanced antenna technologies, OBP, topology-aware routing, and AI integration, creates a resilient infrastructure exceeding the sum of its components.

Future HMLCs will embrace agile design through implicit fitness evaluation frameworks coupling high-fidelity simulation with heuristic optimization, while rapid evaluation models and dynamic adaptation capabilities will enable real-time CON reconfiguration. These developments will transform space-based networks from static infrastructures into intelligent, adaptive systems capable of meeting evolving service requirements across regional, global, and deep space environments.

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